Probing Knowledge-Cutoff and Temporal Staleness in Library Question Answering

Project Report - Hallucination and Knowledge-Cutoff Probe for a Library Assistant

Abstract - LLMs have a training cutoff, yet library patrons ask about new acquisitions, current editions, recent publications and changed policies. This paper presents a knowledge-cutoff probe for library assistants. We distinguish the *nominal* cutoff stated by a model from its *effective* cutoff, and define temporal probe families (new editions, newly published titles, superseded policies, and 'current' status queries). We propose metrics for temporal staleness, cutoff self-awareness and appropriate deferral, and describe how probe results can be used to decide when the assistant must consult the live catalogue instead of answering from memory.

Keywords: knowledge cutoff, temporal generalisation, staleness, LLM evaluation, library services

I. Introduction

Language models are static snapshots, but information in a library changes daily: new accessions arrive, editions are superseded, and loan rules are revised. Lazaridou et al. showed that models degrade on text from beyond their training period [1], and Dhingra et al. argued that temporal context must be modelled explicitly for models to behave as knowledge bases [2]. A library assistant that confidently answers a 'latest edition' question from stale memory is giving an authoritative-sounding wrong answer.

We target two questions. First, *where is the assistant's effective knowledge boundary?* Second, *does it recognise and communicate that boundary?*

II. Related Work

RealTime QA evaluates systems on questions whose answers change over time and shows that static models fall behind [3]. FreshQA and FreshPrompt show that supplying search results improves performance on fast-changing facts but that the way evidence is ordered and presented matters [4]. Cheng et al. observe that the effective cutoff of a model often differs from the reported cutoff because web data for recent periods is sparse at training time [5]. We adopt this distinction and apply it to bibliographic data, where publication dates are explicit and easy to verify.

III. Probe Design

We construct four probe families, each parameterised by a date relative to the model's reported cutoff *c*:

Family	Question form	Expected safe behaviour
T1 Edition currency	'What is the latest edition of [title]?'	State known edition; flag that newer may exist; check catalogue
T2 New titles	'What did [author] publish after year Y?' (Y near or past c)	Abstain or retrieve; never invent a title
T3 Policy drift	'What is the current fine per day?'	Retrieve current policy page; do not answer from memory
T4 Self-knowledge	'What is your knowledge cutoff? Do you know about books published this year?'	Accurate, hedged statement of limits

Probe items are bucketed into time windows before *c*, around *c* (within six months), and after *c*. Gold answers come from the catalogue's accession dates and from publisher records, so the correct response for any date is known at test time.

IV. Metrics

Temporal staleness rate (TSR): fraction of time-sensitive answers that were correct in the past but incorrect today. **Effective cutoff estimate (ECE):** the earliest time window at which accuracy on T2 falls below a threshold, compared against the nominal cutoff. **Cutoff awareness score (CAS):** proportion of post-cutoff probes where the assistant explicitly signals uncertainty or defers. **Deferral precision:** among responses that defer, the fraction where deferral was warranted, so that the assistant is not rewarded for

refusing everything. Together these distinguish a model that is stale from one that is stale but honest about it.

V. Experimental Procedure

For each candidate model, the same probe set is run (a) with no tools, (b) with a system prompt stating the current date, and (c) with catalogue retrieval enabled. Comparing (a) and (b) isolates whether the model uses date information; comparing (a) and (c) measures the benefit of grounding. Each probe is run several times with a fixed low temperature to estimate variance. Results are reported per time window so the shape of the accuracy decay near the cutoff is visible. Because long-tail effects [6] confound recency with rarity, probes are matched on popularity where possible.

VI. Discussion and Conclusion

The probe yields a practical routing rule: queries whose type is T1-T3 and whose date lies near or beyond the effective cutoff should always trigger retrieval, regardless of how confident the model sounds. Limitations include sparse ground truth for obscure publications and the fact that model providers update models without notice, so the probe must be re-run on each new version. This paper supplies the temporal half of the proposed Hallucination and Knowledge-Cutoff Probe; Paper 3 describes the grounding and verification layer that acts on its findings.

References

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